DIRECTIONAL CURVATURE EXPANSION OF THE BAKER CUBIC COEFFICIENT ON RIEMANNIAN SURFACES THROUGH FIFTH ORDER

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ABSTRACT. Let p be a point of a smooth Riemannian surface, let $w \in T_pM$ be a unit bisector direction, and let ρ be the radius of a sufficiently small geodesic circle. A symmetric pair of radial geodesics determines an intrinsic signed side-length law. Encoding an angular error by this side length, multiplying by a scale $q > 1$, and decoding through the inverse side law produces an odd local defect map whose leading term is cubic. We derive the complete directional curvature expansion of its cubic coefficient through order ρ^5. For the scale $q = 4/3$ arising in the Baker approximate-trisection refinement, the result is

$$\begin{aligned} C_3(p,w,\rho) = &-\frac{7}{384} + \frac{7}{384}K\rho^2 + \frac{7}{512}(\nabla_w K)\rho^3 \\ &+ \left[-\frac{7}{1152}K^2 - \frac{7}{7680}\Delta K + \frac{49}{7680}\nabla^2 K(w,w)\right]\rho^4 \\ &+ \left[-\frac{21}{2560}K\nabla_w K - \frac{7}{11520}\nabla_w \Delta K + \frac{49}{23040}\nabla^3 K(w,w,w)\right]\rho^5 + \mathcal{O}(\rho^6). \end{aligned}$$

The proof combines Brewin's corrected degree-six and degree-seven two-point distance expansions with exact surface tensor reduction and a short encode–scale–decode inversion. Orientation reversal removes all mixed components with an odd number of Jw slots, while direction reversal forces the alternating even–odd invariant structure. The coefficients are checked symbolically with exact rational arithmetic, against the exact spherical law and the classical small-circle expansion, and by independent high-precision intrinsic-geodesic computations on a ring torus. Symmetric averages independently recover the quartic coefficient, while four meridional latitudes—including a negative-curvature point—give an overdetermined fifth-order invariant system and numerically recover all three weight-five multipliers. The result is derived from Brewin's published degree-six and degree-seven formulas, which are used as source data and not rederived; Section 9 records independent re-implementations of their surface reduction, a transcription comparison, and an endpoint-swap symmetry test. The normal-coordinate and invariant-theoretic machinery is classical; the contribution is the explicit realization and exact coefficient determination for this intrinsic cubic refinement law.

1. INTRODUCTION

Approximate geometric constructions often admit local error maps whose leading nonlinear coefficient is universal in Euclidean geometry but acquires curvature corrections on a Riemannian surface. The construction considered here originates in a straightedge-and-compass approximate angle-trisection method due to the author and documented by Rostamian [10]. Its Euclidean refinement law has a cubic leading defect. The purpose of this paper is not to revisit exact angle trisection, but to study the intrinsic local coefficient produced when the same encode–scale–decode mechanism is placed on a smooth surface.

The central object is a signed side-length law on a small geodesic circle. Two equal-radius radial directions are placed symmetrically about a unit vector w, with signed angular separation x. Their endpoints have intrinsic distance $c_{\rho,w}(x)$. The map $x \mapsto c_{\rho,w}(x)$ is locally odd and invertible. The encoded side length is multiplied by a fixed scale $q > 1$, decoded through the inverse side law, and

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compared with the original angular error. Oddness forces the calibrated defect map to contain only odd powers of the error variable, so the first nonlinear coefficient is cubic.

The Euclidean coefficient is $-7/384$ at the scale $q = 4/3$. On a sphere of constant positive curvature, the exact coefficient is $-(7/384)\cos^2(\sqrt{K}\,\rho)$. These facts suggest a local curvature expansion, but they do not determine the directional terms on variable-curvature surfaces. The terms through ρ^3 follow from the classical degree-five two-point distance expansion. The present work derives the degree-six and degree-seven contributions and thereby determines the coefficient through ρ^5.

The coefficient is mathematically useful beyond its origin in approximate trisection because a single scalar observable separates successive layers of local surface geometry. The ρ^2 and ρ^3 terms recover K and its directional derivative, while the next two orders resolve the even weight-four basis $\{K^2, \Delta K, \nabla^2 K(w,w)\}$ and the odd weight-five basis $\{K\nabla_w K, \nabla_w \Delta K, \nabla^3 K(w,w,w)\}$. Thus the trisection construction supplies the motivating refinement law, while the Riemannian problem is to determine exactly how that intrinsic law organizes local curvature invariants.

The required two-point geodesic-distance coefficients come from the corrected v3 version of Brewin's normal-coordinate calculation [2]. Brewin's public Cadabra and LaTeX repository provides an implementation-level source for the same formulas [3]. Because higher-order normal-coordinate formulas are convention-sensitive, we first verify that Brewin's lower-order terms reproduce the curvature convention used here. Only then are the degree-six and degree-seven formulas reduced on a surface.

The resulting invariant pattern alternates with the radial order:

order	$w \mapsto -w$	invariants
ρ^2	even	K,
ρ^3	odd	$\nabla_w K$,
ρ^4	even	$K^2,\ \Delta K,\ \nabla^2 K(w,w)$,
ρ^5	odd	$K\nabla_w K,\ \nabla_w \Delta K,\ \nabla^3 K(w,w,w)$.

This is consistent with classical local invariant theory [5, 6]. We do not claim the invariant-weight framework itself as new. The new content is the reduction of the intrinsic encode–scale–decode law to this invariant basis and the exact determination of all coefficient multipliers through radial order five, together with source-locked symbolic and independent numerical checks.

Main result. All curvature quantities below are evaluated at p.

Theorem 1.1 (General scale and Baker specialization)**.** *Let (M, g) be a smooth Riemannian surface, $p \in M$, $w \in T_pM$ a unit vector, and let $\rho > 0$ be sufficiently small that the geodesic circle of radius ρ lies in a convex normal neighbourhood of p. For $q > 1$, let $C_{3,q}(p, w, \rho)$ be the calibrated cubic coefficient defined in Section 2. Then*

$$\begin{aligned} C_{3,q}(p,w,\rho) = \frac{q^2-1}{q^2}\Bigg[&-\frac{1}{24} + \frac{1}{24}K\rho^2 + \frac{1}{32}(\nabla_w K)\rho^3 \\ &+ \left(-\frac{1}{72}K^2 - \frac{1}{480}\Delta K + \frac{7}{480}\nabla^2 K(w,w)\right)\rho^4 \\ &+ \left(-\frac{3}{160}K\nabla_w K - \frac{1}{720}\nabla_w \Delta K + \frac{7}{1440}\nabla^3 K(w,w,w)\right)\rho^5 + \mathcal{O}(\rho^6)\Bigg]. \end{aligned} \tag{1.1}$$

For the Baker scale $q = 4/3$, write $C_3 = C_{3,4/3}$. Then

$$\begin{aligned} C_3(p,w,\rho) = &-\frac{7}{384} + \frac{7}{384}K\rho^2 + \frac{7}{512}(\nabla_w K)\rho^3 \\ &+ \left[-\frac{7}{1152}K^2 - \frac{7}{7680}\Delta K + \frac{49}{7680}\nabla^2 K(w,w)\right]\rho^4 \\ &+ \left[-\frac{21}{2560}K\nabla_w K - \frac{7}{11520}\nabla_w \Delta K + \frac{49}{23040}\nabla^3 K(w,w,w)\right]\rho^5 + \mathcal{O}(\rho^6). \end{aligned} \tag{1.2}$$

The proof occupies Sections 2 to 5. Exact checks and independent numerical validation are presented in Sections 7 and 8. The theorem is proved *relative to* Brewin's published degree-six and degree-seven two-point distance formulas (Appendix A), which are taken as input and are not rederived here; independent checks of their transcription and surface reduction are recorded in Section 9.

2. Intrinsic side law and cubic defect coefficient

Fix a local orientation at p and let $J : T_pM \to T_pM$ be rotation through $\pi/2$. For a small signed angular separation x, put

$$u_x = \cos\frac{x}{2}\, w - \sin\frac{x}{2}\, Jw, \qquad v_x = \cos\frac{x}{2}\, w + \sin\frac{x}{2}\, Jw. \tag{2.1}$$

The vectors u_x and v_x are symmetric about w and have signed angular separation x. For $x \geq 0$, define

$$\widehat{c}_{\rho,w}(x) = d\Big(\exp_p(\rho u_x), \exp_p(\rho v_x)\Big),$$

and define its odd extension by

$$c_{\rho,w}(x) = \begin{cases} \widehat{c}_{\rho,w}(x), & x > 0, \\ 0, & x = 0, \\ -\widehat{c}_{\rho,w}(-x), & x < 0. \end{cases} \tag{2.2}$$

For sufficiently small x, this is a smooth odd local coordinate with positive derivative at the origin.

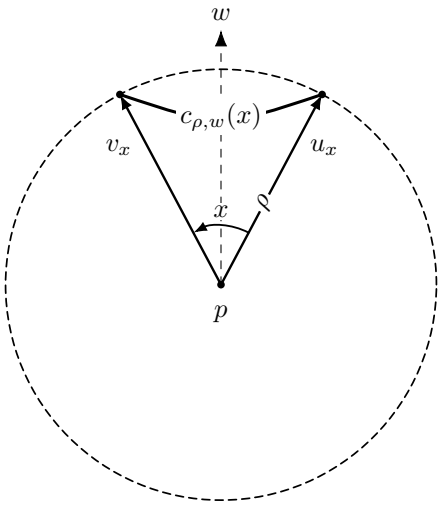


Figure 1. Symmetric equal-radius endpoint geometry. The drawing is schematic; all distances in the definition are intrinsic.

For a scale factor $q > 1$, define the local encode–scale–decode map $\Phi_{\rho,w,q}$ by

$$c_{\rho,w}(\Phi_{\rho,w,q}(x)) = q\, c_{\rho,w}(x), \qquad \Phi_{\rho,w,q}(0) = 0, \tag{2.3}$$

using the branch with positive derivative. The calibrated defect map is

$$F_{\rho,w,q}(e) = e - \Phi_{\rho,w,q}(e/q). \tag{2.4}$$

Oddness of the side law and its inverse implies

$$F_{\rho,w,q}(e) = C_{3,q}(p,w,\rho)e^3 + C_{5,q}(p,w,\rho)e^5 + \cdots . \tag{2.5}$$

Thus even powers of the error variable are structurally forbidden before any extrapolation is applied.

Lemma 2.1 (Coefficient ratio)**.** *If*

$$c_{\rho,w}(x) = b_1(\rho)x + b_3(\rho)x^3 + \mathcal{O}(x^5),$$

then

$$C_{3,q}(p,w,\rho) = \frac{q^2-1}{q^2}\frac{b_3(\rho)}{b_1(\rho)}. \tag{2.6}$$

Proof. Write $\Phi(x) = qx + Ax^3 + \mathcal{O}(x^5)$. Comparing cubic coefficients in $c(\Phi(x)) = qc(x)$ gives

$$b_1 A + b_3 q^3 = q b_3, \qquad A = q(1-q^2)\frac{b_3}{b_1}.$$

Substitution into $F(e) = e - \Phi(e/q)$ yields (2.6). In particular, no x^5 coefficient of the side law is required to determine the cubic defect coefficient. □

3. Two-point distance expansion and convention lock

We use

$$R(A,B)C = \nabla_A\nabla_B C - \nabla_B\nabla_A C - \nabla_{[A,B]}C, \qquad \mathcal{R}(A,B,C,D) = \langle R(A,B)C, D\rangle,$$

so that on a surface

$$\mathcal{R}(A,B,B,A) = K\big(|A|^2|B|^2 - \langle A,B\rangle^2\big). \tag{3.1}$$

Positive Gaussian curvature therefore gives a positive value in the slots (A,B,B,A).

In component notation, the geometric four-slot tensor therefore satisfies

$$\mathcal{R}_{abcd} = K\left(g_{ad}g_{bc} - g_{ac}g_{bd}\right). \tag{3.2}$$

Brewin's displayed arc-length formulas use a component array with the opposite four-slot sign relative to this convention. We therefore define

$$\widetilde{R}_{abcd} := -\mathcal{R}_{abcd} = K\left(g_{ac}g_{bd} - g_{ad}g_{bc}\right). \tag{3.3}$$

All transcriptions of Brewin's Equations 11.21b–e below use the source array $\widetilde{R}$. This distinction is essential because the geometric tensor and Brewin's component array have opposite signs in the relevant slot ordering.

The lower-order two-point expansion may be written

$$\begin{aligned} d(\exp_p X, \exp_p Y)^2 = |X-Y|^2 &- \frac{1}{3}\mathcal{R}(X,Y,Y,X) \\ &- \frac{1}{12}(\nabla_{X+Y}\mathcal{R})(X,Y,Y,X) + \mathcal{O}((|X|+|Y|)^6). \end{aligned} \tag{3.4}$$

This agrees with the distance expansion used by Pennec after conversion to the convention (3.1) [8, Eq. (9)]. For the higher orders we use Brewin's corrected v3 arc-length formulas [2]. The arXiv record notes that v2 corrected computational errors and v3 corrected an index/sign issue in a separate group of displayed formulas; the repository contains the revised Cadabra and LaTeX sources [3].

Write the squared-distance expansion in the form

$$d(\exp_p X, \exp_p Y)^2 = |D|^2 + \sum_{n\geq 4} H_n(X, D), \tag{3.5}$$

where H_n denotes the homogeneous degree-n contribution in (X, D) in Brewin's normal-coordinate expansion.

The remainder after the displayed homogeneous terms admits the following local factorization.

Lemma 3.1 (Uniform normalized remainder)**.** *Fix a point p and a sufficiently small convex normal neighbourhood. Let $D = Y - X$, and write $d^2(\exp_p X, \exp_p Y) = |D|^2 + E(X, D)$. After subtracting the homogeneous terms of total degree at most seven, the remainder along (4.1) can be written*

$$E_{\geq 8}(\rho, x) = 4\rho^2 s(x)^2\, \rho^6 \mathcal{R}(\rho, x, w),$$

where $\mathcal{R}$ is jointly smooth near $(\rho, x) = (0, 0)$, uniformly in unit directions w. In particular its x derivatives of each fixed finite order are bounded locally, so normalized coefficient extraction through radial order five carries a uniform $O(\rho^6)$ remainder. The normalized squared side law is even in x under endpoint interchange.

Proof. In the smooth coordinates (X, D), the function E vanishes at $D = 0$ together with its first derivative in D. Indeed the squared distance and $|D|^2$ both have a zero of order at least two on the diagonal. The iterated integral form of Hadamard's lemma yields

$$E(X, D) = D^a D^b F_{ab}(X, D), \qquad F_{ab}(X, D) = \int_0^1 (1 - t)\, \partial_{D^a} \partial_{D^b} E(X, tD)\, dt,$$

with F_{ab} smooth. Taylor expansion of F_{ab} in the joint variables (X, D) through total degree five has an integral remainder of order six, whose coefficient is smooth jointly in the expansion parameters. Multiplying by $D^a D^b$ gives exactly the omitted total-degree-eight-and-higher terms. Along (4.1), X, D are smooth and $O(\rho)$ and $D = 2\rho s(x) J w$. Hence the Taylor remainder is $4\rho^2 s(x)^2 \rho^6 \mathcal{R}(\rho, x, w)$, with smooth $\mathcal{R}$. Compactness of the unit tangent circle supplies uniform local bounds in w. The interchange of the two endpoints sends x to $-x$, leaving their squared distance unchanged; the factor $4\rho^2 s(x)^2$ is also even. Thus the normalized squared side law is even in x. The locally positive normalized square root is smooth, and differentiating its x-Taylor coefficients preserves the claimed radial remainder. Since $b_1(\rho)/\rho \to 1$, division to form b_3/b_1 also preserves it. □

Brewin writes $x = X$ and $Dx = D = Y - X$. His Equations 11.21b–c are

$$3H_4 = -X^a X^b D^c D^d \widetilde{R}_{acbd}, \tag{3.6}$$

$$12H_5 = -X^a X^b D^c D^d D^e \nabla_c \widetilde{R}_{adbe} - 2X^a X^b X^c D^d D^e \nabla_a \widetilde{R}_{bdce}. \tag{3.7}$$

Lemma 3.2 (Generic convention checksum)**.** *Under (3.1)–(3.3), (3.6)–(3.7) reproduce exactly the two curvature terms in (3.4).*

Proof. Define $\Omega(U, V)^2 = |U|^2|V|^2 - \langle U, V\rangle^2$. Because $D = Y - X$, one has $\Omega(X, D) = \Omega(X, Y)$. On a surface,

$$X^a X^b D^c D^d \widetilde{R}_{acbd} = K\Omega(X, D)^2 = \mathcal{R}(X, Y, Y, X),$$

which proves the H_4 identity. Since $\nabla g = 0$,

$$\nabla_m \widetilde{R}_{abcd} = (\nabla_m K)(g_{ac}g_{bd} - g_{ad}g_{bc}).$$

The two contractions in (3.7) reduce to

$$(\nabla_D K)\Omega(X, D)^2, \qquad (\nabla_X K)\Omega(X, D)^2.$$

Their weighted sum is $(\nabla_D K + 2\nabla_X K)\Omega(X, D)^2 = (\nabla_{X+Y} K)\Omega(X, Y)^2$ because $D + 2X = X + Y$. This gives the H_5 identity. □

This lower-order checksum is performed symbolically for generic vectors before either higher-order formula is used.

4. Surface reduction through homogeneous degree seven

Put

$$s = \sin\frac{x}{2}, \qquad c = \cos\frac{x}{2}, \qquad X = \rho(cw - sJw), \qquad Y = \rho(cw + sJw),$$

so that

$$D = Y - X = 2\rho sJw, \qquad X + Y = 2\rho cw, \qquad |X - Y|^2 = 4\rho^2 s^2. \tag{4.1}$$

Let

$$G_{abcd} = g_{ac}g_{bd} - g_{ad}g_{bc}.$$

The Brewin source array satisfies $\widetilde{R} = KG$, and G is parallel. Hence

$$\nabla\widetilde{R} = (\nabla K)\otimes G, \qquad \nabla^2\widetilde{R} = (\nabla^2 K)\otimes G, \qquad \nabla^3\widetilde{R} = (\nabla^3 K)\otimes G, \tag{4.2}$$

with derivative ordering inherited from Brewin's formula.

4.1. Orders ρ^2 and ρ^3.

From (3.4),

$$c_{\rho,w}(x)^2 = 4\rho^2 s^2\left[1 + A_2\rho^2 + A_3\rho^3 + \mathcal{O}(\rho^4)\right], \tag{4.3}$$

where

$$A_2 = -\frac{1}{3}Kc^2, \qquad A_3 = -\frac{1}{6}(\nabla_w K)c^3. \tag{4.4}$$

The first term is isotropic; the first directional term is odd under $w \mapsto -w$.

4.2. Homogeneous degree six.

Choose the orthonormal frame $e_1 = w$, $e_2 = Jw$ and write

$$H_{11} = \nabla^2 K(w,w), \quad H_{12} = \nabla^2 K(w,Jw), \quad H_{22} = \nabla^2 K(Jw,Jw).$$

Brewin's Equation 11.21d contains three polynomial blocks. Exact contraction of those blocks gives

$$\begin{aligned} T_{1,R} &= -128K^2c^2s^4, & T_{1,H} &= -96H_{22}c^2s^4,\\ T_{2,R} &= -128K^2c^2s^4, & T_{2,H} &= 144c^2s^3(H_{22}s - H_{12}c),\\ T_{3,R} &= 64K^2c^2s^2(c^2+s^2), & T_{3,H} &= -72c^2s^2(H_{22}s^2 - 2H_{12}cs + H_{11}c^2). \end{aligned}$$

The mixed-Hessian terms are $-144H_{12}c^3s^3$ and $+144H_{12}c^3s^3$, and therefore cancel exactly.

Proposition 4.1 (Degree-six distance coefficient)**.** *The homogeneous degree-six term is*

$$H_6 = \frac{\rho^6 c^2 s^2}{45}\left[8K^2c^2 - 24K^2s^2 - 9H_{11}c^2 - 3H_{22}s^2\right]. \tag{4.5}$$

Consequently,

$$A_4 = \frac{c^2}{180}\left[8K^2c^2 - 24K^2s^2 - 9H_{11}c^2 - 3H_{22}s^2\right]. \tag{4.6}$$

Proof. Sum the six displayed contractions and use $c^2 + s^2 = 1$. Division by the leading factor $4\rho^2 s^2$ in the squared side law gives (4.6). □

The cancellation of H_{12} is also forced geometrically: under $J \mapsto -J$ the two endpoint rays exchange while their distance is unchanged, but $\nabla^2 K(w,Jw)$ changes sign.

4.3. **Homogeneous degree seven.** Let $e_0 = w$, $e_1 = Jw$, and put

$$K_w = \nabla_w K, \qquad K_J = \nabla_{Jw} K, \qquad T_{abc} = \nabla_{e_a}\nabla_{e_b}\mathrm{d}K(e_c),$$

with derivative ordering matching Brewin's Equation 11.21e. The complete source transcription is recorded in the accompanying audit file; its four blocks contain $\widetilde{R}\nabla\widetilde{R}$ and $\nabla^3\widetilde{R}$ terms. After exact surface contraction, every component with an odd number of Jw slots cancels.

For direct auditability, the eight individual contractions entering Brewin's four degree-seven blocks are listed below. These are the quantities in $1080H_7/\rho^7$ before summation:

$$\begin{aligned}
B_{1,R\nabla R} &= 32Kc^2s^4(6K_Js - 17K_wc),\\
B_{1,\nabla^3 R} &= 64c^2s^4[-(T_{011}+T_{101}+T_{110})c + 3T_{111}s],\\
B_{2,R\nabla R} &= -576KK_Jc^2s^5,\\
B_{2,\nabla^3 R} &= -96T_{111}c^2s^5,\\
B_{3,R\nabla R} &= 192Kc^2s^3[K_Jc^2 + 3K_Js^2 - 2K_wcs],\\
B_{3,\nabla^3 R} &= -48c^2s^3[T_{001}c^2 + T_{010}c^2 - 2T_{011}cs\\
&\qquad + T_{100}c^2 - 2T_{101}cs - 2T_{110}cs + 3T_{111}s^2],\\
B_{4,R\nabla R} &= -192Kc^2s^2(c^2+s^2)(K_Js - K_wc),\\
B_{4,\nabla^3 R} &= 48c^2s^2[-T_{000}c^3 + (T_{001}+T_{010}+T_{100})c^2s\\
&\qquad - (T_{011}+T_{101}+T_{110})cs^2 + T_{111}s^3].
\end{aligned}$$

Their sum gives (4.7); in particular, all K_J terms and all third-derivative components with an odd number of Jw slots cancel.

Proposition 4.2 (Degree-seven distance coefficient)**.** *The homogeneous degree-seven term satisfies*

$$\begin{aligned}
1080\frac{H_7}{\rho^7} = &-16c^3s^2[-12KK_wc^2 + 46KK_ws^2 + 3T_{000}c^2\\
&+ (T_{011}+T_{101}+T_{110})s^2].
\end{aligned} \tag{4.7}$$

The coefficients of

$$K_J, \quad T_{001}, \quad T_{010}, \quad T_{100}, \quad T_{111}$$

vanish identically.

Proof. The four $\widetilde{R}\nabla\widetilde{R}$ blocks and four $\nabla^3\widetilde{R}$ blocks are contracted with $X/\rho = (c, -s)$ and $D/\rho = (0, 2s)$ in the frame (w, Jw). Summation gives (4.7). The vanished components are exactly those that are odd under orientation reversal $J \mapsto -J$. □

Dividing by $4\rho^7 s^2$ gives

$$A_5 = \frac{c^3}{270}\left[(12c^2 - 46s^2)KK_w - 3c^2T_{000} - s^2(T_{011}+T_{101}+T_{110})\right]. \tag{4.8}$$

Remark 4.3 (Third-derivative ordering independence). Although the intermediate tensor components inherit Brewin's ordered covariant-derivative convention, the fifth-order result depends on them only through T_{000} and the symmetric sum $T_{011} + T_{101} + T_{110}$. Any global permutation of the three derivative slots fixes T_{000} and merely permutes the three terms in this sum. Consequently the final scalar coefficient in (5.13) is independent of the chosen third-derivative slot-ordering convention.

5. Side-law extraction and proof of the theorem

Write the squared side law through degree seven as

$$c_{\rho,w}(x)^2 = 4\rho^2 s^2 \left[1 + A_2\rho^2 + A_3\rho^3 + A_4\rho^4 + A_5\rho^5 + \mathcal{O}(\rho^6)\right]. \tag{5.1}$$

The signed square root is

$$\begin{aligned} c_{\rho,w}(x) = 2\rho s\Bigg[1 + \frac{1}{2}A_2\rho^2 + \frac{1}{2}A_3\rho^3 + \left(\frac{1}{2}A_4 - \frac{1}{8}A_2^2\right)\rho^4 \\ + \left(\frac{1}{2}A_5 - \frac{1}{4}A_2A_3\right)\rho^5 + \mathcal{O}(\rho^6)\Bigg]. \end{aligned} \tag{5.2}$$

The cross terms $-A_2^2/8$ and $-A_2A_3/4$ are essential.

Define

$$B_4 = \frac{1}{2}A_4 - \frac{1}{8}A_2^2, \qquad B_5 = \frac{1}{2}A_5 - \frac{1}{4}A_2A_3.$$

Then

$$B_4 = \frac{c^2}{120}\left[(1-9s^2)K^2 - 3c^2H_{11} - s^2H_{22}\right], \tag{5.3}$$

$$B_5 = \frac{c^3}{1080}\left[(9c^2 - 92s^2)KK_w - 6c^2T_{000} - 2s^2(T_{011}+T_{101}+T_{110})\right]. \tag{5.4}$$

Write

$$\begin{aligned} b_1(\rho) &= \rho\left[1 + p_2\rho^2 + p_3\rho^3 + p_4\rho^4 + p_5\rho^5 + \mathcal{O}(\rho^6)\right], \\ b_3(\rho) &= \rho\left[-\frac{1}{24} + r_2\rho^2 + r_3\rho^3 + r_4\rho^4 + r_5\rho^5 + \mathcal{O}(\rho^6)\right]. \end{aligned}$$

Expanding (5.2) in x gives

$$p_2 = -\frac{1}{6}K, \qquad r_2 = \frac{7}{144}K, \tag{5.5}$$

$$p_3 = -\frac{1}{12}K_w, \qquad r_3 = \frac{5}{144}K_w, \tag{5.6}$$

$$p_4 = \frac{K^2}{120} - \frac{H_{11}}{40}, \qquad r_4 = -\frac{61}{2880}K^2 + \frac{13}{960}H_{11} - \frac{1}{480}H_{22}, \tag{5.7}$$

$$p_5 = \frac{3KK_w - 2T_{000}}{360}, \qquad r_5 = -\frac{58KK_w - 8T_{000} + T_{011} + T_{101} + T_{110}}{2160}. \tag{5.8}$$

At fourth order, direct series division yields

$$\left[\rho^4\right]\frac{b_3}{b_1} = -\frac{1}{72}K^2 - \frac{1}{480}\Delta K + \frac{7}{480}\nabla^2 K(w,w). \tag{5.9}$$

At fifth order,

$$\left[\rho^5\right]\frac{b_3}{b_1} = -\frac{77KK_w - 15T_{000} + 2T_{011} + 2T_{101} + 2T_{110}}{4320}. \tag{5.10}$$

The ordered third derivatives are converted to invariant form by the following commutator identity.

Lemma 5.1 (Ordered third derivatives)**.** *At p, choose the frame (w, Jw) normal at p. Then*

$$T_{101} = T_{110} = T_{011} + K\nabla_w K, \qquad \nabla_w \Delta K = T_{000} + T_{011}. \tag{5.11}$$

Proof. Hessian symmetry gives $T_{110} = T_{101}$. For a covector α,

$$([\nabla_a, \nabla_b]\alpha)_c = -R^d{}_{cab}\alpha_d.$$

Taking $\alpha = \mathrm{d}K$, $a = 1$, $b = 0$, and $c = 1$ in the convention (3.1) gives $T_{101} - T_{011} = K\nabla_w K$. The trace identity in a normal frame gives $\nabla_w \Delta K = T_{000} + T_{011}$. □

Therefore

$$T_{011} + T_{101} + T_{110} = 3\nabla_w \Delta K - 3T_{000} + 2K\nabla_w K. \tag{5.12}$$

Substituting (5.12) into (5.10) gives

$$\left[\rho^5\right] \frac{b_3}{b_1} = -\frac{3}{160} K\nabla_w K - \frac{1}{720}\nabla_w \Delta K + \frac{7}{1440}\nabla^3 K(w, w, w). \tag{5.13}$$

Combining the constant, quadratic, cubic, quartic, and quintic ratio coefficients with Lemma 2.1 proves Theorem 1.1.

6. Symmetry and invariant structure

Proposition 6.1 (Alternating parity)**.** *The coefficient of ρ^m in $C_{3,q}(p, w, \rho)$ is even in w for even m and odd in w for odd m, through the orders computed here.*

Proof. Formally extend the radial parameter to signed ρ. Since $u_x(-w) = -u_x(w)$ and $v_x(-w) = -v_x(w)$, the endpoint pair for $(-w, \rho)$ is exactly the endpoint pair for $(w, -\rho)$. Hence the signed side laws, and therefore the cubic coefficients, satisfy

$$C_{3,q}(p, -w, \rho) = C_{3,q}(p, w, -\rho)$$

as formal power series in ρ. Comparing coefficients gives

$$[\rho^m]C_{3,q}(p, -w, \rho) = (-1)^m[\rho^m]C_{3,q}(p, w, \rho),$$

which proves the stated alternating parity. The geometric radius remains positive; signed ρ is used only as a formal device for the local series. Equivalently, the statement can be read at the level of the squared side law $c_{\rho,w}(x)^2 = 4\rho^2 s^2[1 + \sum_m A_m \rho^m]$: its coefficients A_m are determined by the smooth function $\rho \mapsto d^2(\exp_p(\rho u_x), \exp_p(\rho v_x))$, which is defined for signed ρ, and the ratio b_3/b_1 is unchanged by the overall sign of $2\rho s$. □

Remark 6.2 (Orientation reversal)*.* The auxiliary operator J is not part of the final data. Under $J \mapsto -J$, the two endpoint rays exchange, so the side length is unchanged. Consequently every surviving scalar contains an even number of Jw slots. The exact cancellations in Propositions 4.1 and 4.2 are the tensor-level realization of this symmetry.

Corollary 6.3 (Invariant bases through weight five)**.** *The nonconstant coefficients through order ρ^5 lie in*

$$\begin{aligned}
&\rho^2 : \operatorname{span}\{K\},\\
&\rho^3 : \operatorname{span}\{\nabla_w K\},\\
&\rho^4 : \operatorname{span}\{K^2, \Delta K, \nabla^2 K(w, w)\},\\
&\rho^5 : \operatorname{span}\{K\nabla_w K, \nabla_w \Delta K, \nabla^3 K(w, w, w)\}.
\end{aligned}$$

7. Analytic checks

7.1. **Flat and constant-curvature geometries.** If the metric is flat near p, all curvature corrections vanish and

$$C_3 = -\frac{7}{384}.$$

For positive constant curvature K, spherical trigonometry gives the exact signed side law

$$c_\rho(x) = \frac{2}{\sqrt{K}} \arcsin\left(\sin(\sqrt{K}\rho)\sin\frac{x}{2}\right).$$

Its coefficient ratio is

$$\frac{b_3}{b_1} = -\frac{1}{24}\cos^2(\sqrt{K}\rho),$$

so

$$C_3 = -\frac{7}{384}\cos^2(\sqrt{K}\rho). \tag{7.1}$$

Since

$$\cos^2\alpha = 1 - \alpha^2 + \frac{1}{3}\alpha^4 - \frac{2}{45}\alpha^6 + \cdots,$$

(7.1) confirms the $K\rho^2$ and $K^2\rho^4$ coefficients and the absence of odd powers for constant curvature.

7.2. **Classical geodesic-circle circumference.** The coefficient $b_1(\rho)$ is the geodesic-circle line element at the endpoint in direction w. From (5.5)–(5.8),

$$\frac{b_1}{\rho} = 1 - \frac{1}{6}K\rho^2 - \frac{1}{12}(\nabla_w K)\rho^3 + \left(\frac{K^2}{120} - \frac{1}{40}\nabla^2 K(w,w)\right)\rho^4 + \left(\frac{K\nabla_w K}{120} - \frac{1}{180}\nabla^3 K(w,w,w)\right)\rho^5 + \mathcal{O}(\rho^6).$$

Averaging over the unit circle in T_pM removes the odd directional terms and uses

$$\frac{1}{2\pi}\int \nabla^2 K(w,w)\,\mathrm{d}w = \frac{1}{2}\Delta K.$$

Thus the geodesic-circle circumference is

$$L_p(\rho) = 2\pi\rho\left[1 - \frac{1}{6}K\rho^2 + \left(\frac{K^2}{120} - \frac{\Delta K}{80}\right)\rho^4 + \mathcal{O}(\rho^6)\right]. \tag{7.2}$$

This agrees with the Bertrand–Diguet–Puiseux and modern small-ball expansions [1, 4, 9, 7]. It also explains why the odd directional corrections are invisible in the total circumference.

7.3. **Ring-torus exact specializations.** For convenience, write

$$Q_m(p,w) := [\rho^m]C_3(p,w,\rho).$$

For the ring torus

$$ds^2 = \mathrm{d}\theta^2 + (3+\cos\theta)^2\mathrm{d}\phi^2, \qquad K(\theta) = \frac{\cos\theta}{3+\cos\theta},$$

the quartic formula reproduces the exact rational anchors in Table 1.

For this $R = 3$, $r = 1$ ring torus, the exact identity

$$Q_{4,\mathrm{in},\theta} - Q_{4,\mathrm{in},\phi} = -4\,(Q_{4,\mathrm{out},\theta} - Q_{4,\mathrm{out},\phi}) = \frac{49}{10240}$$

follows immediately.

At $\theta = \pi/2$ with $w = e_\theta$,

$$K = 0, \qquad \nabla_w K = -\frac{1}{3}, \qquad \nabla_w \Delta K = \frac{2}{9}, \qquad \nabla^3 K(w,w,w) = \frac{1}{9}.$$

TABLE 1. Exact order-ρ^4 torus specializations of (1.2). These values illustrate the formula and are not independent numerical validations.

branch	K	ΔK	$\nabla^2 K(w,w)$	Q_4
zero-curvature azimuthal	0	$-1/9$	$1/9$	$7/8640$
outer azimuthal	$1/4$	$-3/16$	0	$-77/368640$
outer meridional	$1/4$	$-3/16$	$-3/16$	$-259/184320$
inner azimuthal	$-1/2$	$3/4$	0	$-203/92160$
inner meridional	$-1/2$	$3/4$	$3/4$	$119/46080$

Therefore

$$Q_3 = -\frac{7}{1536}, \qquad Q_5 = \frac{7}{69120}. \tag{7.3}$$

8. Independent high-precision validation

The numerical calculations are not used in the proof. They supply an independent metric-level audit in which intrinsic torus geodesic distances are computed without evaluating the normal-coordinate formulas. The same data provide two complementary tests: the symmetric average of opposite bisector directions isolates the even radial orders and independently checks Q_4, while the antisymmetric average isolates the odd orders and checks Q_5. Four meridional latitudes, including the negative-curvature point $\theta = 2\pi/3$, make the fifth-order invariant recovery overdetermined.

8.1. **Independent quartic check from symmetric averages.**

For opposite meridional directions $w_+ = e_\theta$ and $w_- = -e_\theta$, define

$$S(\rho) = \frac{C_3(p, w_+, \rho) + C_3(p, w_-, \rho)}{2}.$$

By the parity relation of Proposition 6.1,

$$S(\rho) = Q_0 + Q_2\rho^2 + Q_4\rho^4 + \mathcal{O}(\rho^6).$$

Using the same seven radii as the fifth-order calculation, an unrestricted even fit through ρ^8 recovers the quartic coefficient without fixing Q_0 or Q_2 in advance. The fitted quartic coefficients are presented in Table 2. The corresponding relative differences are 2.44×10^{-11}, 4.49×10^{-11}, 8.47×10^{-11}, and 6.48×10^{-10}, respectively. Thus the quartic coefficient is independently checked on both positive- and negative-curvature meridional configurations.

TABLE 2. Independent quartic amplitudes from the symmetric average. The fitted value comes from an unrestricted five-term even fit $S(\rho) = c_0 + c_2\rho^2 + c_4\rho^4 + c_6\rho^6 + c_8\rho^8$.

θ	K	predicted Q_4	fitted c_4
$\pi/4$	0.1907435698305462	$-1.414108712929600 \times 10^{-3}$	$-1.414108712895040 \times 10^{-3}$
$\pi/3$	$1/7$	$-1.415462018140590 \times 10^{-3}$	$-1.415462018077032 \times 10^{-3}$
$\pi/2$	0	$-1.316550925925926 \times 10^{-3}$	$-1.316550925814465 \times 10^{-3}$
$2\pi/3$	$-1/5$	$-6.368055555555556 \times 10^{-4}$	$-6.368055559681878 \times 10^{-4}$

8.2. **Zero-curvature fifth-order audit.** For the fifth-order test, let $w_+ = e_\theta$ and $w_- = -e_\theta$ at the zero-curvature latitude. Direction antisymmetrization gives

$$A(\rho) = \frac{C_3(p, w_+, \rho) - C_3(p, w_-, \rho)}{2} = -\frac{7}{1536}\rho^3 + \frac{7}{69120}\rho^5 + \mathcal{O}(\rho^7).$$

Hence

$$E_5(\rho) = \frac{A(\rho) + \frac{7}{1536}\rho^3}{\rho^5} = \frac{7}{69120} + \mathcal{O}(\rho^2). \tag{8.1}$$

The independent solver uses unit speed and Clairaut's first integral for radial geodesics, a separate turning-point shooting problem for the joining geodesic, arbitrary-precision quadrature, and nested polynomial fits of the side law. The cubic-subtracted residual has global exponent

$$4.998860308456601429,$$

and the even-power extrapolation

$$E_5(\rho) = a_0 + a_2\rho^2 + a_4\rho^4$$

gives

$$a_0 = 0.00010127314814679543194502656 5,$$

with relative difference 1.3357×10^{-11} from $7/69120$. Including an $a_6\rho^6$ term reduces the relative difference to 4.4445×10^{-14}.

More importantly, the raw residual obeys the predicted functional form. Across a threefold range in radius, the ratio of relative error to ρ^2 is essentially constant: This verifies not merely a close

TABLE 3. Predicted $\mathcal{O}(\rho^2)$ residual in the normalized fifth-order amplitude.

ρ	relative error in $E_5(\rho)$	relative error$/\rho^2$
0.020	1.612×10^{-4}	0.4030
0.030	3.627×10^{-4}	0.4030
0.060	1.449×10^{-3}	0.4026

decimal value but the predicted residual law.

The maximum nested-window spread in C_3 was approximately 2.58×10^{-25}, the largest turning-point shooting residual was below 9.84×10^{-49}, and independent precision/quadrature and angular-grid audits reproduced the extrapolated amplitude. Thus the zero-curvature fifth-order combination is independently validated on this torus configuration.

8.3. **Overdetermined four-latitude fifth-order audit.** To test the $K\nabla_w K$ multiplier and to introduce a negative-curvature configuration, the same intrinsic-geodesic calculation was run at $\theta = \pi/4$, $\pi/3$, $\pi/2$, and $2\pi/3$, again with $w = e_\theta$ and its reverse. Each run used 60-decimal arithmetic with 48-node Gauss–Legendre quadrature and an independent 70-decimal, 64-node precision/quadrature audit. All polynomial least-squares fits were solved by high-precision QR factorization rather than normal equations. A target-free odd fit

$$A(\rho) = a_3\rho^3 + a_5\rho^5 + a_7\rho^7 + a_9\rho^9$$

was used to extract the fifth-order amplitude a_5 without fixing the cubic coefficient.

The relative errors in these four unrestricted fifth-order amplitudes are respectively 3.80×10^{-11}, 8.01×10^{-11}, 9.05×10^{-11}, and 4.25×10^{-10}. These per-latitude comparisons are the direct numerical tests of the analytic coefficient.

For the same four latitudes, let the rows of the 4×3 matrix M be

$$(K\nabla_w K,\ \nabla_w \Delta K,\ \nabla^3 K(w, w, w)).$$

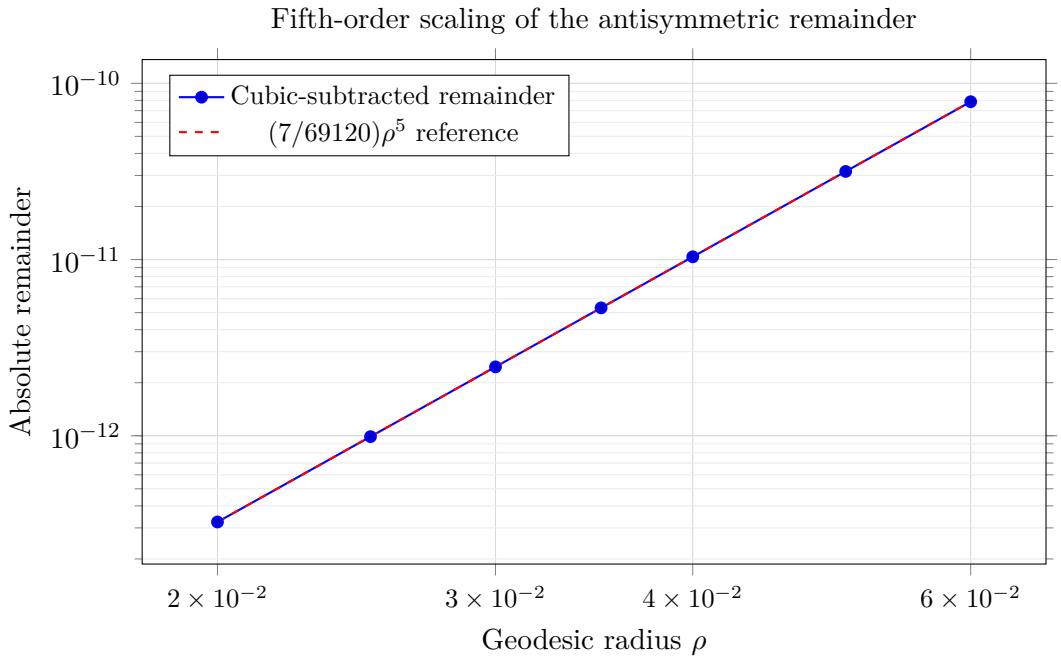


FIGURE 2. Fifth-order scaling of the antisymmetric remainder after subtraction of the known cubic term.

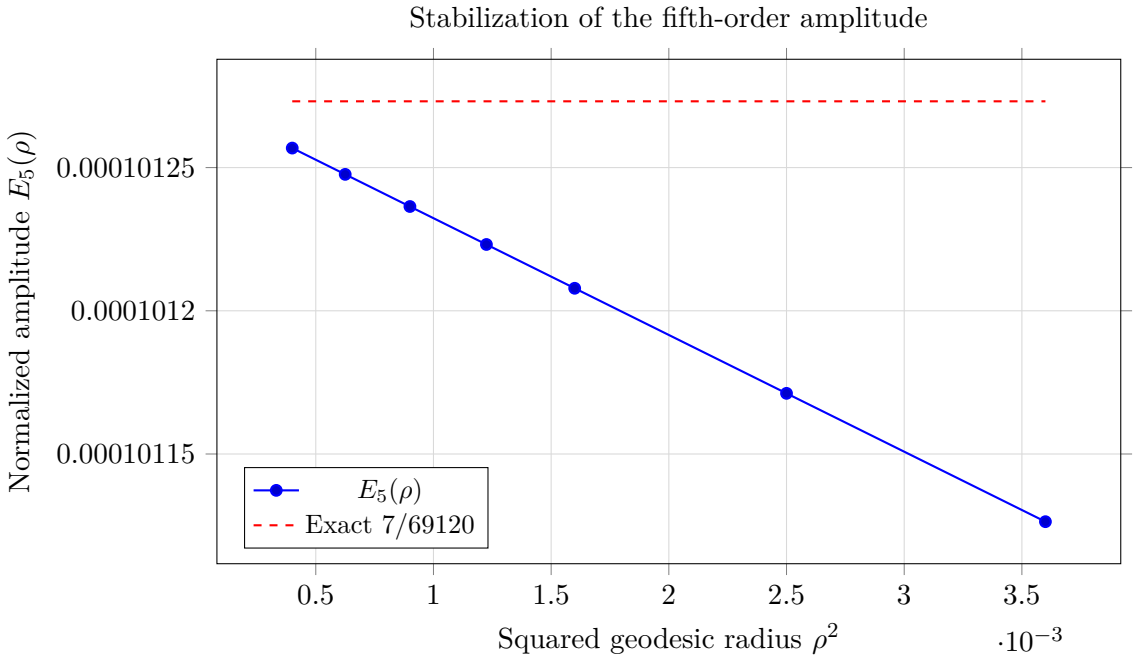


FIGURE 3. Stabilization of the normalized fifth-order amplitude toward the exact target 7/69120.

The matrix has rank three. A high-precision QR least-squares solve of the overdetermined system $M\beta = a_5$ gives

$$\beta = (-0.00820312499900897,\ -0.000607638888658034,\ 0.00212673611057080),$$

TABLE 4. Independent fifth-order meridional amplitudes from four torus latitudes. The numerical column is the unrestricted fitted a_5 from the 70-decimal audit.

θ	K	predicted Q_5	unrestricted fitted a_5
$\pi/4$	0.1907435698305462	$1.104442495077554 \times 10^{-4}$	$1.104442495119574 \times 10^{-4}$
$\pi/3$	$1/7$	$1.157225393783871 \times 10^{-4}$	$1.157225393876608 \times 10^{-4}$
$\pi/2$	0	$1.012731481481481 \times 10^{-4}$	$1.012731481389811 \times 10^{-4}$
$2\pi/3$	$-1/5$	$2.626943724812797 \times 10^{-4}$	$2.626943723695210 \times 10^{-4}$

for the multipliers of $K\nabla_w K$, $\nabla_w \Delta K$, and $\nabla^3 K(w,w,w)$, respectively. The exact analytic values are

$$\left(-\frac{21}{2560}, -\frac{7}{11520}, \frac{49}{23040}\right),$$

and the corresponding relative differences are 1.21×10^{-10}, 3.80×10^{-10}, and 2.54×10^{-10}. The residual norm is 2.18×10^{-15}, or 6.73×10^{-12} relative to the norm of the measured amplitude vector; the numerical 2-norm condition number of M is approximately 86.76. Thus the fourth latitude supplies genuine redundancy: the recovered multipliers are consistent with all four independently fitted amplitudes rather than being forced by a square three-by-three solve.

Scope of the numerical tests. All torus runs use meridional directions $w = \pm e_\theta$ on a single ring torus ($R = 3$, $r = 1$). On this torus $\nabla^2 K(w,w) = K''$ and $\nabla^2 K(Jw, Jw) = (f'/f)K'$ vary independently along the meridian, so the quartic comparison does probe the two Hessian multipliers separately. No azimuthal run ($w = e_\phi$) has been performed at order ρ^5; such a run would give an additional test of the Jw-orientation structure. The invariant basis used in the four-latitude recovery is justified analytically (Section 6); the numerics confirm the multipliers within that basis.

9. Independent verification of the symbolic chain

This section records checks performed on October 25, 2026 in an AI-assisted review. They are independent re-implementations and consistency tests of the *symbolic* chain; none of them rederives Brewin's formulas, and the original symbolic review did not rerun the torus geodesic computations of Section 8; the subsequent independent $\theta = \pi/3$ run is distinguished in item (g). The single script `curvature_v24_checks.py` reproduces items (a), (b), (d), (e), and (g) below; item (f) refers to a separate script.

(a) **Side-law extraction.** Starting from the squared side law (5.1) with $A_2, \ldots, A_5$ as displayed, the coefficients $p_2, \ldots, p_5$ and $r_2, \ldots, r_5$ of (5.5)–(5.8), the ratios (5.9) and (5.10), the invariant form (5.13), and all nine multipliers of (1.2) were recomputed in exact arithmetic and agree.

(b) **Surface contraction of Brewin's formulas.** The degree-six and degree-seven expressions of Appendix A were contracted on a surface in a second, independent implementation (and again in a third, written from Brewin's published equations with all third-derivative components T_{abc} treated as independent). All three reproduce Propositions 4.1 and 4.2 identically, including the cancellation of H_{12} and of the coefficients of $K_J, T_{001}, T_{010}, T_{100}, T_{111}$.

(c) **Comparison with Brewin's printed text (externally reported).** An independent reviewer reports comparing Eqs. 11.21b–e of arXiv:0903.2087v3 (revised November 4, 2022) with (3.6), (3.7), and the expressions in Appendix A: 1, 2, 7, and 17 terms respectively, 27 in all. Coefficients, index placement, derivative-slot order, and block structure reportedly agree, including the previously uninspected tail of Eq. 11.21e. The reviewer reports seven intentional single-term corruptions were all detected. This comparison concerns Brewin's *printed arXiv text*, not an independent derivation or comparison with his Cadabra source. The comparison program `brewin_1121_transcription_check.py` was described by the reviewer but was not supplied for local execution or inclusion in the present archive. Accordingly, this source-comparison

result is recorded as externally reported, not locally reproduced. (The sign-convention lock $\widetilde{R} = -\mathcal{R}$ is checked separately in Section 9.)

(d) **Endpoint-swap symmetry.** Let $P = \exp_p X$ and $Q = \exp_p Y$. Interchanging P and Q replaces (X, D) by $(X + D, -D)$, preserves homogeneous degree, and leaves d^2 unchanged; hence every homogeneous term satisfies $H_n(X, D) = H_n(X + D, -D)$. After surface reduction (using the relations of Lemma 5.1), both H_6 and H_7 as transcribed in Appendix A satisfy this identity for generic X, D. The test has power: changing any single coefficient of H_7 by one unit destroys the symmetry, with the exception of the $+10$ term of the first block, which agrees with Brewin's printed text according to the comparison reported in item (c). The symmetry thus independently constrains the same coefficients that item (c) compares with the printed text. The test is insensitive to the sign convention of the commutator in Lemma 5.1; that sign is instead fixed by item (e) and by the four-latitude fits. The symmetry is a necessary consistency condition on the source formulas, not a derivation of them.

(e) **Ordered third derivatives.** The relations (5.11), in particular $T_{101} - T_{011} = K\nabla_w K$, $T_{110} = T_{101}$ and $\nabla_w \Delta K = T_{000} + T_{011}$, were verified directly from the metric of the ring torus at general θ, together with the exact anchors $Q_3 = -7/1536$, $Q_5 = 7/69120$ and the five quartic anchors of Table 1.

(f) **Coefficient chain (separate audit).** An independent script (`curvature_coefficient_chain_audit.py`) takes the surface-reduced coefficients as inputs and verifies the quotient b_3/b_1 through ρ^5, the general-q factor, the $q = 4/3$ specialization, and the constant-curvature check; it agrees with item (a). The companion closeout archive now includes this script and its checksum.

(g) **Tabulated numerics (from reported values).** The predicted quartic and quintic amplitudes at the four latitudes, the quoted relative differences, the residual-law ratios, and the 4×3 least-squares recovery of the weight-five multipliers (including the residual norm 2.18×10^{-15} and the condition number 86.76) were recomputed from the fitted amplitudes *as reported* in Section 8 and agree. The underlying four-latitude fits were not regenerated in this original audit. A subsequent independent run regenerated the $\theta = \pi/3$ meridional fit at 60-digit working precision using 48-node quadrature, returning an unrestricted fifth-order amplitude $a_5 = 1.157225393876608 \times 10^{-4}$ to the displayed digits. The remaining three latitude fits have not been independently rerun in that subsequent check.

TABLE 5. Summary of the independent verification layers and what each does and does not establish.

layer	establishes	does not establish
(a), (f) extraction	algebra from A_m to (1.2)	values of A_4, A_5
(b) contractions	surface reduction of Appendix A	that Appendix A matches Brewin (see (c))
(c) text comparison	printed 11.21b–e: 27 terms (reported)	correctness of Brewin; Cadabra comparison
(d) swap symmetry	coefficients consistent with d^2 symmetric	derivation of Brewin's formula
(e) torus identities	commutator lemma, anchors	–
(g) numerical checks	reported tables; $\pi/3$ fit rerun	other three fits rerun

10. Reproducibility, scope, and limitations

The verification record identifies the following scripts (items 6–9 were added in version 2.4). The companion closeout archive includes the recovered scripts and numerical packages identified in its checksum manifest; scripts absent from that archive are historical references and are not asserted to be reproduced by this release:

(1) `verify_brewin_degree6_surface_reduction.py`, which performs the generic lower-order convention lock, encodes Brewin 11.21d, reduces the degree-six tensors, and checks all quartic coefficients and torus anchors;
(2) `verify_brewin_degree7_surface_reduction.py`, which repeats the convention lock, encodes Brewin 11.21e, reduces the degree-seven tensors, verifies orientation cancellation, and derives the fifth-order invariant coefficient;
(3) `audit_ordered_derivatives_torus.py`, which constructs the torus Christoffel symbols, Hessian, and ordered third derivatives from the metric and checks the commutator identities for general θ;
(4) `rho5_torus_high_precision_validation.py`, which performs the original zero-curvature high-precision geodesic calculation;
(5) `rho5_torus_multilatitude_validation.py`, which generalizes the intrinsic-geodesic calculation to an arbitrary meridional base latitude and is used here at $\theta = \pi/4$, $\pi/3$, $\pi/2$, and $2\pi/3$; it also extracts Q_4 from symmetric averages and uses QR-based polynomial least squares;
(6) `curvature_independent_audit.py`, an independent contraction audit written from Brewin's published Eqs. 11.21d–e, with arbitrary independent third-derivative components;
(7) `curvature_coefficient_chain_audit.py`, the independent coefficient-chain audit of Section 9(f);
(8) `curvature_v24_checks.py`, which reproduces the checks of Section 9 (a), (b), (d), (e), and (g); and
(9) `brewin_1121_transcription_check.py`, an externally reported term-by-term source-comparison checker (not supplied or executed in this archive).

The supplied symbolic verification scripts use exact arithmetic. The separately archived numerical reproduction records side samples, fitted coefficients, diagnostics, environment metadata, and checksums. The public project repository is https://github.com/CWBaker360/curvature-deformed-arcsine. The companion closeout archive supplies SHA-256 checksums of its included files. Historical scripts not in that archive are not included in the accompanying evidence; no GitHub release tag is claimed.

The result is local in ρ and in the error variable e. No global convergence or large-radius assertion is made.

Higher-dimensional outlook. The same encode–scale–decode construction can be formulated on a higher-dimensional Riemannian manifold (M, g) after choosing an oriented orthonormal two-frame (w, v) in T_pM. This two-frame determines the oriented tangent plane

$$\Pi = \operatorname{span}\{w, v\},$$

which contains the symmetric endpoint rays. At leading curvature order, one naturally expects the Gaussian curvature appearing in the surface formula to be replaced by the sectional curvature $K_p(\Pi)$ of this plane.

At higher orders, however, the coefficients need not reduce to scalar combinations of a single curvature function. They may involve contractions of the full Riemann tensor $\mathcal{R}$, its covariant derivatives $\nabla^{(k)}\mathcal{R}$, the metric g, and the distinguished vectors w and v. Consequently, the resulting invariant basis may depend not only on the plane Π, but also on the distinguished directions within that plane.

Unlike the surface case, a single unit vector $w \in T_pM$ does not determine a canonical rotation operator on the full tangent space T_pM. Once an oriented two-plane Π has been chosen, there is a canonical quarter-turn

$$J_\Pi \colon \Pi \longrightarrow \Pi, \qquad J_\Pi w = v, \qquad J_\Pi v = -w,$$

but this operator has no canonical extension to all of T_pM. An auxiliary choice of $v \perp w$, or equivalently of an oriented orthonormal two-frame, is therefore required to define the symmetric endpoint geometry.

A higher-dimensional classification would consequently require both a larger invariant basis and a precise account of the resulting two-frame dependence. Establishing such a classification is a natural extension of the present work, but lies outside its scope. The surface result obtained here provides a foundational model for understanding how local geometric invariants are organized within the Baker cubic refinement coefficient.

The normal-coordinate expansion and natural-tensor organization are classical. The result establishes the explicit coefficients for this particular intrinsic refinement law and does not claim a new general theory of local invariants.

11. Conclusion

The cubic coefficient of the intrinsic encode–scale–decode law separates successive layers of local geometry. The universal Euclidean coefficient is followed by Gaussian curvature, its directional derivative, an even weight-four combination of quadratic curvature and Hessian data, and an odd weight-five combination of $K\nabla_w K$, $\nabla_w \Delta K$, and $\nabla^3 K(w,w,w)$. The coefficients are derived analytically from source-locked normal-coordinate formulas, verified by exact symbolic computation, and checked against classical and constant-curvature geometry. Independent high-precision intrinsic-geodesic computations recover the quartic coefficient from symmetric averages and, across four torus latitudes including $K<0$, give an overdetermined fifth-order test that numerically recovers all three weight-five invariant multipliers.

For the Baker scale $q=4/3$, the complete result through fifth order is (1.2). This gives an explicit curvature extension of the Euclidean cubic refinement law, derived from Brewin's published two-point distance expansion and checked as described in Sections 8 and 9, and explicitly identifies the successive local curvature invariants encoded by its cubic coefficient. The derivation is conditional on those source formulas, whose transcription has been compared with the published text (Section 9(c)) but not rederived.

AI-assistance and provenance disclosure

AI tools were used in reviewing this manuscript and in the independent verification of Section 9: re-implementing the surface contractions, re-deriving the side-law extraction, testing the endpoint-swap symmetry, and recomputing tabulated quantities from reported values. The original torus geodesic computations of Section 8 are author-side results; a subsequent independent review regenerated the single $\theta=\pi/3$ fit, but not the complete four-latitude dataset. The author retains responsibility for the stated mathematical claims.

Acknowledgment

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Appendix A. Source formulas used for the higher-order distance terms

For transparency, this appendix records the two higher-order Brewin inputs in the notation $x = X$, $Dx = D = Y - X$, using the source array $\widetilde{R}$ defined in (3.3). These formulas are not rederived here; their role is as source data for the exact surface reductions, and their transcription is independently audited in the accompanying archive.

The degree-six term is

$$\begin{aligned}
360H_6 = {} & X^aX^bD^cD^dD^eD^f\left(-8g^{gh}\widetilde{R}_{acdg}\widetilde{R}_{befh} - 6\nabla_{cd}\widetilde{R}_{aebf}\right)\\
& + X^aX^bX^cD^dD^eD^f\left(16g^{gh}\widetilde{R}_{adbg}\widetilde{R}_{cefh} - 9\nabla_{da}\widetilde{R}_{becf} - 9\nabla_{ad}\widetilde{R}_{becf}\right)\\
& + X^aX^bX^cX^dD^eD^f\left(16g^{gh}\widetilde{R}_{aebg}\widetilde{R}_{cfdh} - 18\nabla_{ab}\widetilde{R}_{cedf}\right).
\end{aligned}$$

The degree-seven term is

$$\begin{aligned}
1080H_7 = {} & X^aX^bX^cD^dD^eD^fD^g(-4g^{hi}\widetilde{R}_{adeh}\nabla_f\widetilde{R}_{bgci} - 24g^{hi}\widetilde{R}_{adeh}\nabla_b\widetilde{R}_{cfgi}\\
& + 10g^{hi}\widetilde{R}_{adeh}\nabla_i\widetilde{R}_{bfcg} + 16g^{hi}\widetilde{R}_{adbh}\nabla_e\widetilde{R}_{cfgi} - 4\nabla_{dea}\widetilde{R}_{bfcg}\\
& - 4\nabla_{dae}\widetilde{R}_{bfcg} - 4\nabla_{ade}\widetilde{R}_{bfcg})\\
& + X^aX^bD^cD^dD^eD^fD^g(-18g^{hi}\widetilde{R}_{acdh}\nabla_e\widetilde{R}_{bfgi} - 3\nabla_{cde}\widetilde{R}_{afbg})\\
& + X^aX^bX^cX^dD^eD^fD^g(24g^{hi}\widetilde{R}_{aefh}\nabla_b\widetilde{R}_{cgdi} + 24g^{hi}\widetilde{R}_{aebh}\nabla_f\widetilde{R}_{cgdi}\\
& + 24g^{hi}\widetilde{R}_{aebh}\nabla_c\widetilde{R}_{dfgi} - 6\nabla_{eab}\widetilde{R}_{cfdg} - 6\nabla_{aeb}\widetilde{R}_{cfdg} - 6\nabla_{abe}\widetilde{R}_{cfdg})\\
& + X^aX^bX^cX^dX^eD^fD^g(48g^{hi}\widetilde{R}_{afbh}\nabla_c\widetilde{R}_{dgei} - 12\nabla_{abc}\widetilde{R}_{dfeg}).
\end{aligned}$$

Appendix B. Reproducibility commands

From the package root, the exact symbolic checks may be run with

```
python scripts/verify_brewin_degree6_surface_reduction.py
python scripts/verify_brewin_degree7_surface_reduction.py
python scripts/audit_ordered_derivatives_torus.py
```

The high-precision numerical validation may be reproduced with

```
python scripts/rho5_torus_high_precision_validation.py \
  --output-dir outputs/recomputed --precision-audit
```

The nonzero-curvature fifth-order audits may be reproduced with

```
python scripts/rho5_torus_multilatitude_validation.py --u0 pi/4 \
  --output-dir outputs/pi4 --precision-audit
python scripts/rho5_torus_multilatitude_validation.py --u0 pi/3 \
  --output-dir outputs/pi3 --precision-audit
python scripts/rho5_torus_multilatitude_validation.py --u0 2pi/3 \
```

`--output-dir outputs/2pi3 --precision-audit`

The numerical runs are substantially more expensive than the exact symbolic checks.

Appendix C. Revision history

Version 2.3 (September 29, 2026): complete directional curvature expansion through ρ^5; Brewin-based derivation; torus validations; four-latitude fifth-order audit.

Version 2.4 (October 25, 2026): (1) wording of the rigor claim: the theorem is stated and described as proved relative to Brewin's published formulas (Theorem remark, abstract, Conclusion); (2) new Section 9 recording independent verification layers and their limits, with companion script `curvature_v24_checks.py`; (3) externally reported term-by-term comparison of Brewin's printed Eqs. 11.21b–e (27 terms) with Appendix A, including the previously uninspected tail of 11.21e; (4) endpoint-swap symmetry of H_6, H_7 as a consistency test; (5) clarification of the uniformity of the $\mathcal{O}(\rho^6)$ remainder and of the parity argument via the squared side law; (6) explicit statement of the scope of the numerical tests (meridional directions on one torus); (7) archive scripts extended and a provenance flag added; (8) AI-assistance disclosure and this history. No equation, coefficient, or numerical value of v2.3 was changed.

October 29, 2026, version 2.5: Added Lemma 3.1 replacing the standing regularity assumption by a smooth Hadamard-factorization argument; recorded the independent $\theta = \pi/3$ fit reproduction. No coefficient in the main theorem was altered.

Independent researcher